



The extended Kalman filter (EKF)

- EKF makes two simplifying assumptions when applying general sequential inference equations to a nonlinear system:
 1. When computing estimates of the output of a nonlinear function, EKF assumes $\mathbb{E}[\text{fn}(x)] \approx \text{fn}(\mathbb{E}[x])$, which is not true in general.
 2. When computing covariance estimates, EKF uses Taylor-series expansion to linearize the system equations around the present operating point.
- In this lesson, you will learn how to apply these approximations and assumptions to derive the EKF prediction steps.
- In the next lesson, you will learn how to derive the EKF update steps.



EKF step 1a: State prediction time update

- The state prediction step is approximated as:

$$\begin{aligned}\hat{x}_k^- &= \mathbb{E}[f(x_{k-1}, u_{k-1}, w_{k-1}) \mid \mathbb{Z}_{k-1}] \\ &\approx f(\hat{x}_{k-1}^+, u_{k-1}, \bar{w}_{k-1}),\end{aligned}$$

where $\bar{w}_{k-1} = \mathbb{E}[w_{k-1}]$. (Often, $\bar{w}_{k-1} = 0$.)

- That is, we approximate the expected value of the new state by assuming that it is reasonable to simply propagate $\hat{x}_{k-1}^+$ and $\bar{w}_{k-1}$ through the state equation.



EKF step 1b: Error covariance time update (1)

- We start with an expression for $\tilde{x}_k^-$:

$$\tilde{x}_k^- = x_k - \hat{x}_k^- = f(x_{k-1}, u_{k-1}, w_{k-1}) - f(\hat{x}_{k-1}^+, u_{k-1}, \bar{w}_{k-1}).$$

- Approx. first term via Taylor series around prior operating point $\{\hat{x}_{k-1}^+, u_{k-1}, \bar{w}_{k-1}\}$:

$$\begin{aligned}x_k &\approx f(\hat{x}_{k-1}^+, u_{k-1}, \bar{w}_{k-1}) + \underbrace{\frac{df(x_{k-1}, u_{k-1}, w_{k-1})}{dx_{k-1}} \bigg|_{x_{k-1}=\hat{x}_{k-1}^+}}_{\text{Defined as } \hat{A}_{k-1}} (x_{k-1} - \hat{x}_{k-1}^+) \\ &\quad + \underbrace{\frac{df(x_{k-1}, u_{k-1}, w_{k-1})}{dw_{k-1}} \bigg|_{w_{k-1}=\bar{w}_{k-1}}}_{\text{Defined as } \hat{B}_{k-1}} (w_{k-1} - \bar{w}_{k-1}).\end{aligned}$$

- This gives $\tilde{x}_k^- \approx (\hat{A}_{k-1} \tilde{x}_{k-1}^+ + \hat{B}_{k-1} \tilde{w}_{k-1})$.



EKF step 1b: Error covariance time update (2)

- Substituting this to find the prediction-error covariance:
 $\Sigma_{\tilde{x},k}^- = \mathbb{E}[(\tilde{x}_k^-)(\tilde{x}_k^-)^T] \approx \hat{A}_{k-1} \Sigma_{\tilde{x},k-1}^+ \hat{A}_{k-1}^T + \hat{B}_{k-1} \Sigma_{\tilde{w}} \hat{B}_{k-1}^T.$
- Note, by the chain rule of total differentials,

$$df(x_{k-1}, u_{k-1}, w_{k-1}) = \frac{\partial f(x_{k-1}, u_{k-1}, w_{k-1})}{\partial x_{k-1}} dx_{k-1} + \frac{\partial f(x_{k-1}, u_{k-1}, w_{k-1})}{\partial u_{k-1}} du_{k-1} + \frac{\partial f(x_{k-1}, u_{k-1}, w_{k-1})}{\partial w_{k-1}} dw_{k-1}$$

$$\frac{df(x_{k-1}, u_{k-1}, w_{k-1})}{dx_{k-1}} = \frac{\partial f(x_{k-1}, u_{k-1}, w_{k-1})}{\partial x_{k-1}} + \frac{\partial f(x_{k-1}, u_{k-1}, w_{k-1})}{\partial u_{k-1}} \frac{du_{k-1}}{dx_{k-1}} + \frac{\partial f(x_{k-1}, u_{k-1}, w_{k-1})}{\partial w_{k-1}} \frac{dw_{k-1}}{dx_{k-1}} = \frac{\partial f(x_{k-1}, u_{k-1}, w_{k-1})}{\partial x_{k-1}}.$$



EKF step 1c: Output estimate

- Similarly, $\frac{df(x_{k-1}, u_{k-1}, w_{k-1})}{dw_{k-1}} = \frac{\partial f(x_{k-1}, u_{k-1}, w_{k-1})}{\partial w_{k-1}}.$
- Distinction between total and partial differential not yet critical.
- Will be critical in week four of this course.
- System output is estimated to be:

$$\hat{z}_k = \mathbb{E}[h(x_k, u_k, v_k) | \mathbb{Z}_{k-1}] \approx h(\hat{x}_k^-, u_k, \bar{v}_k),$$

where $\bar{v}_k = \mathbb{E}[v_k].$

- That is, it is assumed that propagating $\hat{x}_k^-$ and the mean sensor noise is the best approximation to estimating the output.



Summary

- EKF makes two fundamental assumptions to generalize KF.
 - Assume $\mathbb{E}[\text{fn}(x)] \approx \text{fn}(\mathbb{E}[x])$, which is not true in general.
 - Assume Taylor-series expansion to linearize system equations for covariances.
- Applying these to generic-probabilistic-inference solution yields first three EKF steps (final three in next lesson).
- In general must be careful to distinguish between total and partial differentials (although not critical here).
- First half of EKF is now derived.